SIR SYED UNIVERSITY OF ENGINEERING & TECHNOLOGY
COMPUTER SCIENCE & INFORMATION TECHNOLOGY DEPARTMENT

Fall 2025
IT Project Management (IT-316)
Assignment # 3

Semester: IV — Batch: 2022F
Announced Date: **12 January 2026** — Due Date: **16th January 2026**
Total Marks:3 — Marks Obtained:
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Roll#: 2022F-BCS-152
Section: E

CLO #	Course Learning Outcomes (CLOs)	PLO Mapping	Bloom's Taxonomy
CLO 3	**Apply** project management concepts to explain a given scenario.	PLO-3 (Problem Analysis)	C3 (Applying)

Q1. An IT infrastructure upgrade project is at the 50% completion mark. The following data is recorded:

- **Budget at Completion (BAC):** $800,000
- **Earned Value (EV):** $350,000
- **Planned Value (PV):** $400,000
- **Actual Cost (AC):** $420,000
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Management predicts project performance will worsen due to shortages in skilled personnel:

- **Expected CPI deterioration:** 20%
- **Expected SPI deterioration:** 25%

Using the data provided above:

1. Compute the current **CPI**.
2. Compute the current **SPI**.
3. Calculate the **Adjusted CPI** and **Adjusted SPI** using the deterioration percentages.
4. Using the adjusted CPI, compute the **revised Estimate at Completion (EAC)** with the formula:

$$EAC = AC + \frac{(BAC - EV)}{\text{Adjusted CPI}}$$

5. The original planned duration for the project was **12 months**. Calculate the **revised expected project duration** using:

$$\text{Revised Duration} = \frac{12}{\text{Adjusted SPI}}$$

Provide full calculations and a short interpretation of the impact on cost and schedule.

1) compute the current CPI

$$CPI = \frac{EV}{AC}$$

$$CPI = \frac{350\,000}{420\,000} = 0.833$$

The $CPI < 1$ indicates the project is over budget, spending more than the value earned.

2) Compute the current SPI

$$SPI = \frac{EV}{PV}$$

$$SPI = \frac{350\,000}{400\,000} = 0.875$$

The $SPI < 1$ indicates the project is behind schedule, earning less value than planned

3) Calculate the Adjusted CPI and Adjusted SPI using the deterioration percentages.

The project may slow down due to a shortage of skilled staff so we adjust the CPI and SPI by lowering them according to the expected performance drop

$$\text{Adjusted CPI} = CPI \times (1 - \text{CPI deterioration})$$
$$= 0.833 \times (1 - 0.20)$$
$$= 0.666$$

$$\text{Adjusted SPI} = \text{SPI} \times (1 - \text{SPI deterioration})$$
$$= 0.875 \times (1 - 0.25)$$
$$= 0.656$$

4) Using the adjusted CPI, compute the revised EAC with formulae

$$EAC = AC + \frac{(BAC - EV)}{\text{Adjusted CPI}}$$

$$= 420000 + \frac{800000 - 350000}{0.666}$$

$$= 1095675.676 \approx 1095676$$

The revised EAC of 1095676 indicates the project is expected to exceed the original budget significantly due to performance deterioration.

5) The original planned duration for the project was 12 months. Calculate the revised expected project duration using:

$$\text{Revised Duration} = \frac{12}{\text{Adjusted SPI}}$$

$$\text{Revised Duration} = \frac{12}{0.656} = 18.29$$

The project duration is expected to increase from 12 months to approximately 18.3 months, indicating a major schedule delay.

The analysis reveals that the IT infrastructure upgrade project is currently over budget (CPI < 1) and schedule (SPI < 1).

Considering the expected deterioration in performance due to skilled personnel shortages, the cost overrun could reach approximately $1095676, which is about 37% above the original budget and the project duration may extend to 18.3 months, an increase over 6 months.

Immediate corrections and corrective measures or actions, such as resource allocation, training or hiring additional skilled personnel may be necessary to mitigate cost and schedule risks.